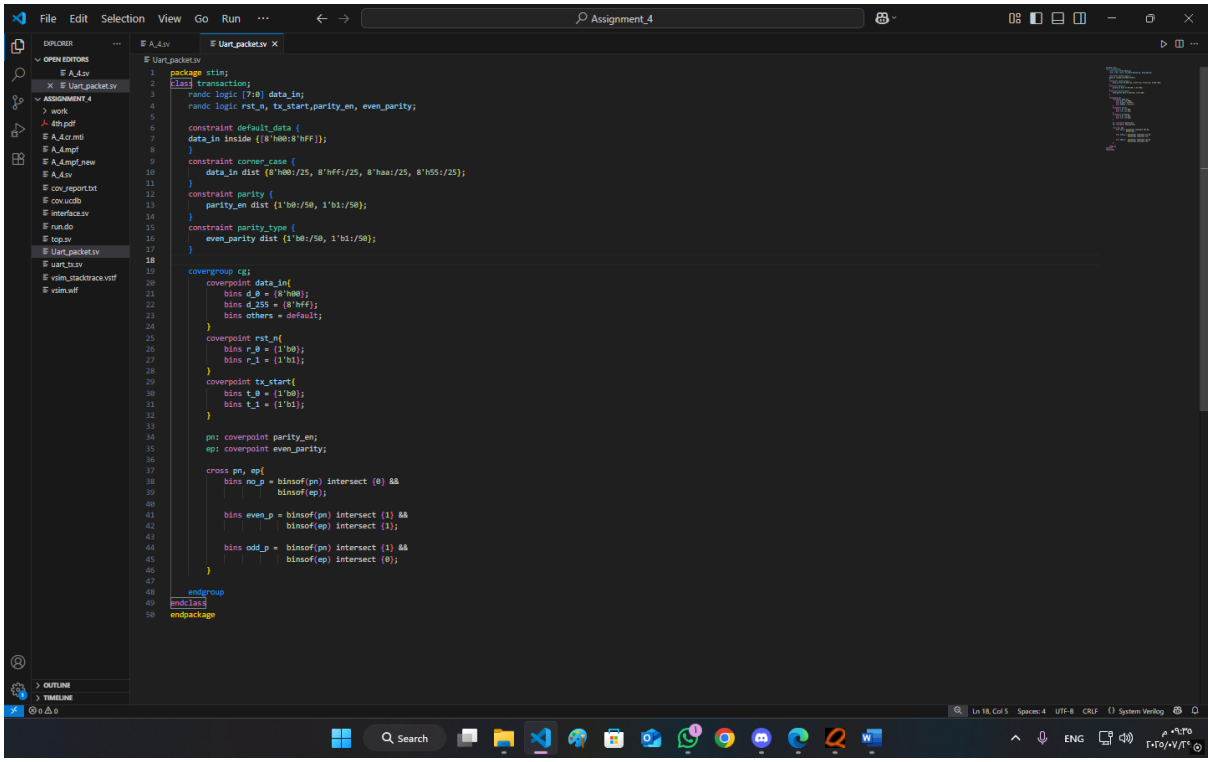
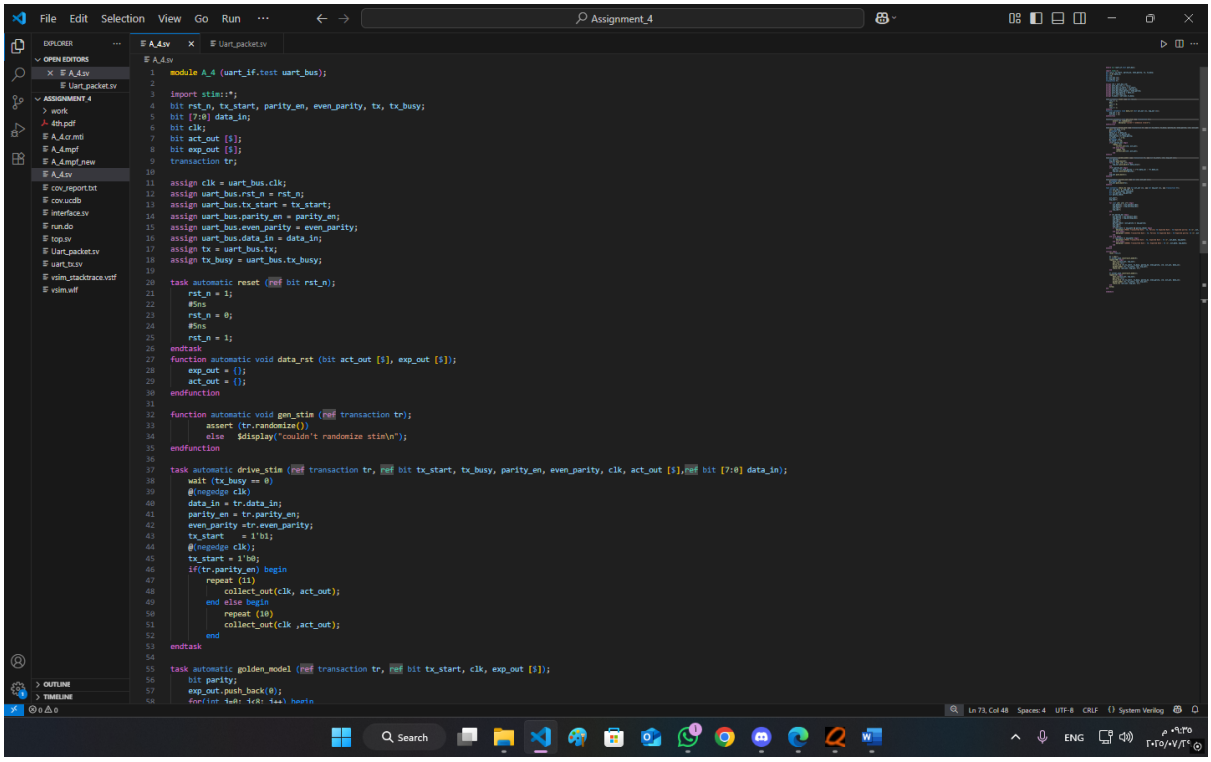
Code snippets:



```
package stim;
class transaction;
    randc logic [7:0] data_in;
    randc logic rst_n, tx_start,parity_en, even_parity;

    constraint default_data {
    data_in inside {[8'h00:8'hFF]};
    }
    constraint corner_case {
        data_in dist {8'h00:/25, 8'hff:/25, 8'haa:/25, 8'h55:/25};
    }
    constraint parity {
        parity_en dist {1'b0:/50, 1'b1:/50};
    }
    constraint parity_type {
        even_parity dist {1'b0:/50, 1'b1:/50};
    }

    covergroup cg;
        coverpoint data_in{
            bins d_0 = {8'h00};
            bins d_255 = {8'hff};
            bins others = default;
        }
        coverpoint rst_n{
            bins r_0 = {1'b0};
            bins r_1 = {1'b1};
        }
        coverpoint tx_start{
            bins t_0 = {1'b0};
            bins t_1 = {1'b1};
        }

        pn: coverpoint parity_en;
        ep: coverpoint even_parity;

        cross pn, ep{
            bins no_p = binsof(pn) intersect {0} &&
                        binsof(ep);

            bins even_p = binsof(pn) intersect {1} &&
                          binsof(ep) intersect {1};

            bins odd_p =  binsof(pn) intersect {1} &&
                          binsof(ep) intersect {0};
        }

    endgroup
endclass
endpackage
```

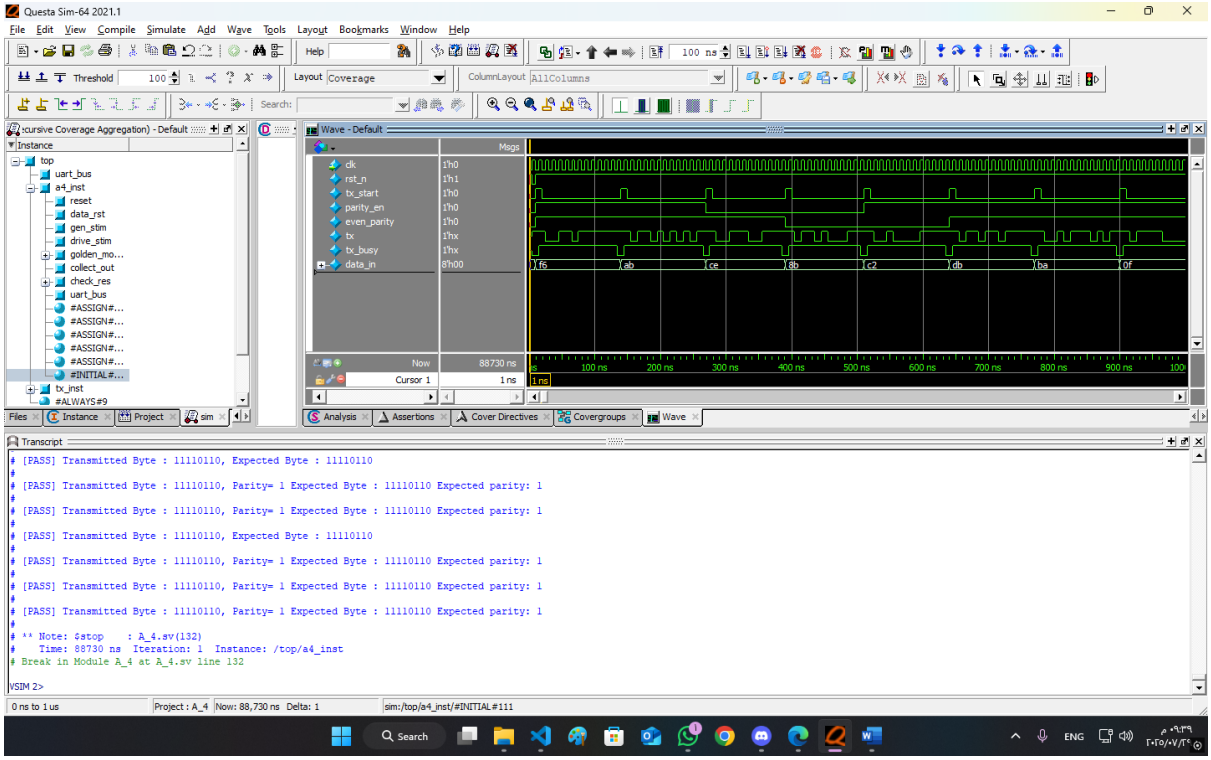


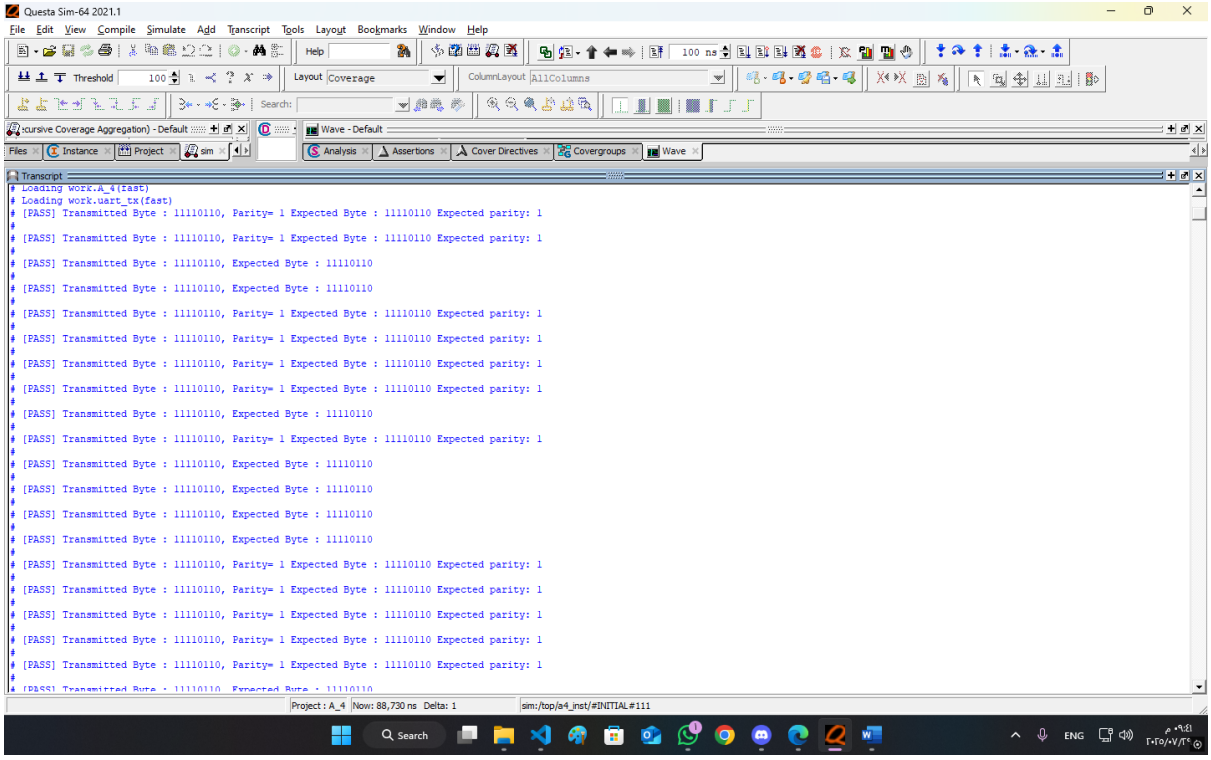
```
module A_4 (uart_if.test uart_bus);

import stim::*;
bit rst_n, tx_start, parity_en, even_parity, tx, tx_busy;
bit [7:0] data_in;
bit clk;
bit act_out [$];
bit exp_out [$];
transaction tr;

assign clk = uart_bus.clk;
assign uart_bus.rst_n = rst_n;
assign uart_bus.tx_start = tx_start;
assign uart_bus.parity_en = parity_en;
assign uart_bus.even_parity = even_parity;
assign uart_bus.data_in = data_in;
assign tx = uart_bus.tx;
assign tx_busy = uart_bus.tx_busy;

task automatic reset (ref bit rst_n);
    rst_n = 1;
    #5ns
    rst_n = 0;
    #5ns
    rst_n = 1;
endtask
function automatic void data_rst (bit act_out [$], exp_out [$]);
    exp_out = {};
    act_out = {};
endfunction

function automatic void gen_stim (ref transaction tr);
        assert (tr.randomize())
        else   $display("couldn't randomize stim\n");
endfunction

task automatic drive_stim (ref transaction tr, ref bit tx_start, tx_busy, parity_en, even_parity, clk, act_out [$],ref bit [7:0] data_in);
    wait (tx_busy == 0)
    @(negedge clk)
    data_in = tr.data_in;
    parity_en = tr.parity_en;
    even_parity =tr.even_parity;
    tx_start    = 1'b1;
    @(negedge clk);
    tx_start = 1'b0;
    if(tr.parity_en) begin
        repeat (11)
            collect_out(clk, act_out);
        end else begin
            repeat (10)
            collect_out(clk ,act_out);
        end
endtask

task automatic golden_model (ref transaction tr, ref bit tx_start, clk, exp_out [$]);
    bit parity;
    exp_out.push_back(0);
    for(int i=0; i<8; i++) begin
```

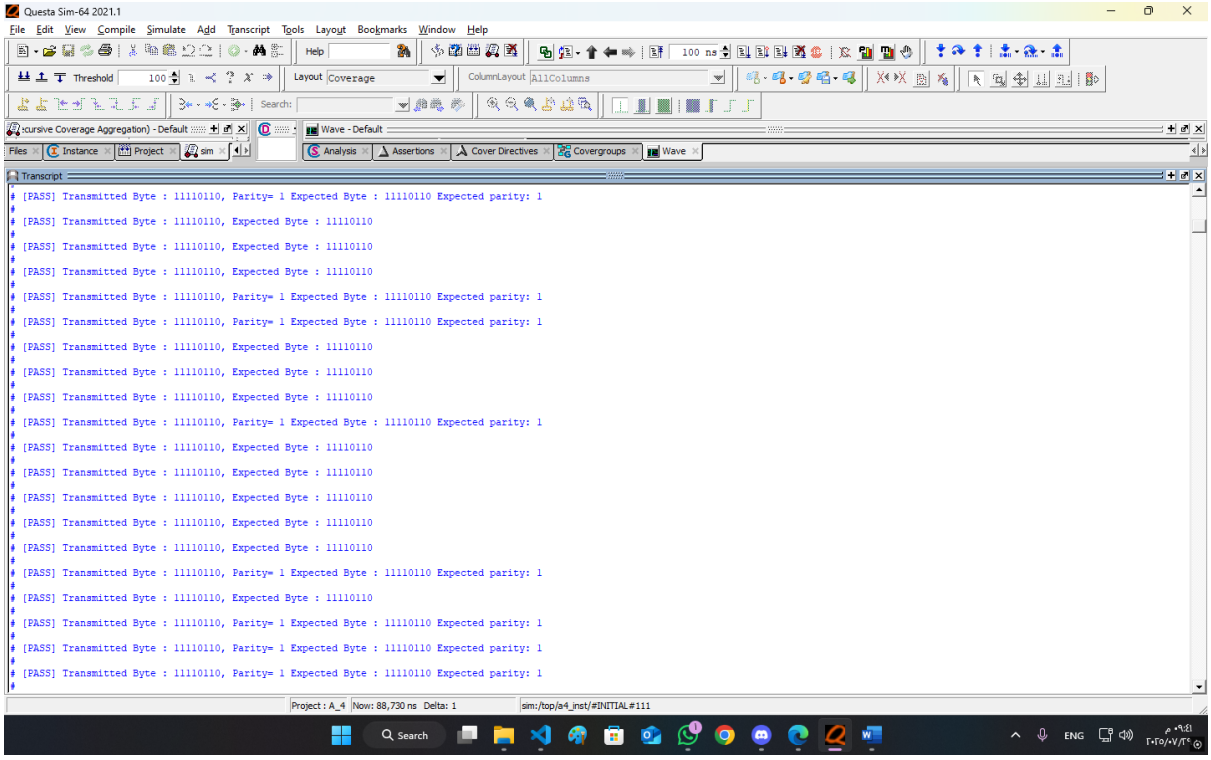
```
task automatic golden_model (ref transaction tr, ref bit tx_start, clk, exp_out [$]);
    for(int j=0; j<8; j++) begin
    end
    if(tr.parity_en) begin
        parity = tr.even_parity ? ~(^tr.data_in) : ^tr.data_in;
        exp_out.push_back(parity);
    end
    exp_out.push_back(1);
endtask

task automatic collect_out (ref bit clk, act_out [$]);
    @(negedge clk)
    act_out.push_back(tx);
endtask

task automatic check_res (ref bit act_out [$], ref bit exp_out [$], ref transaction tr);
    int act_idx = 0, exp_idx = 0;
    bit [7:0] act_byte, exp_byte;
    bit act_parity, exp_parity;
    bit parity_check;

    act_idx++;
    exp_idx++;

    for (int j=0; j<8; j++) begin
        act_byte[j] = act_out[act_idx];
        exp_byte[j] = exp_out[exp_idx];
        act_idx++;
        exp_idx++;
    end

    if (tr.parity_en) begin
        act_parity = act_out[act_idx];
        exp_parity = exp_out[exp_idx];
        act_idx++;
        exp_idx++;
        parity_check = act_parity == exp_parity;
        act_idx++;
        exp_idx++;
        if (act_byte == exp_byte && parity_check) begin
            $display("[PASS] Transmitted Byte : %b, Parity= %b Expected Byte : %b Expected parity: %b \n", act_byte, act_parity, exp_byte, exp_parity);
        end else begin
            $display("[ERROR] Transmitted Byte : %b, Parity= %b Expected Byte : %b Expected parity: %b \n", act_byte, act_parity, exp_byte, exp_parity);
        end
    end else begin
        if (act_byte == exp_byte) begin
            $display("[PASS] Transmitted Byte : %b, Expected Byte : %b \n", act_byte, exp_byte);
        end else begin
            $display("[ERROR] Transmitted Byte : %b, Expected Byte : %b \n", act_byte, exp_byte);
        end
    end
endtask

initial begin
    reset (rst_n);

    tr = new();
```

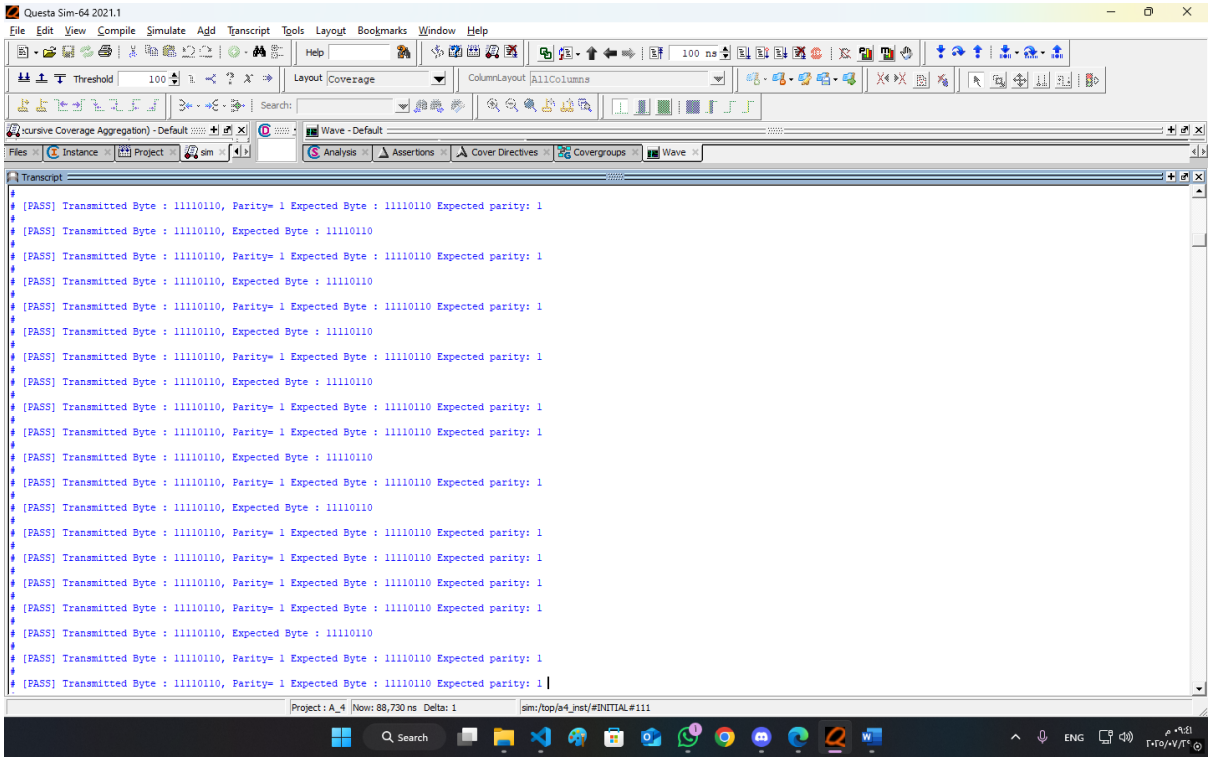
```
task automatic check_res (ref bit act_out [$], ref bit exp_out [$], ref transaction tr);
    for (int j=0; j<8; j++) begin
        act_byte[j] = act_out[act_idx];
        exp_byte[j] = exp_out[exp_idx];
        act_idx++;
        exp_idx++;
    end

    if (tr.parity_en) begin
        act_parity = act_out[act_idx];
        exp_parity = exp_out[exp_idx];
        act_idx++;
        exp_idx++;
        parity_check = act_parity == exp_parity;
        act_idx++;
        exp_idx++;
        if (act_byte == exp_byte && parity_check) begin
            $display("[PASS] Transmitted Byte : %b, Parity= %b Expected Byte : %b Expected parity: %b \n", act_byte, act_parity, exp_byte, exp_parity);
        end else begin
            $display("[ERROR] Transmitted Byte : %b, Parity= %b Expected Byte : %b Expected parity: %b \n", act_byte, act_parity, exp_byte, exp_parity);
        end
    end else begin
        if (act_byte == exp_byte) begin
            $display("[PASS] Transmitted Byte : %b, Expected Byte : %b \n", act_byte, exp_byte);
        end else begin
            $display("[ERROR] Transmitted Byte : %b, Expected Byte : %b \n", act_byte, exp_byte);
        end
    end
endtask

initial begin
    reset (rst_n);

    tr = new();
    tr.corner_case.constraint_mode(0);
    repeat(700) begin
        data_rst(act_out, exp_out);
        gen_stim (tr);
        tr.cg.sample();
        drive_stim (tr,tx_start, tx_busy, parity_en, even_parity, clk, act_out, data_in);
        golden_model (tr, tx_start, clk, exp_out);
        check_res (act_out, exp_out, tr);
    end

    tr.corner_case.constraint_mode(1);
    repeat(10) begin
        data_rst(act_out, exp_out);
        gen_stim (tr);
        tr.cg.sample();
        drive_stim (tr,tx_start, tx_busy, parity_en, even_parity, clk, act_out, data_in);
        golden_model (tr, tx_start, clk, exp_out);
        check_res (act_out, exp_out, tr);
    end
    $stop;
end

endmodule
```

Wave form and logs:



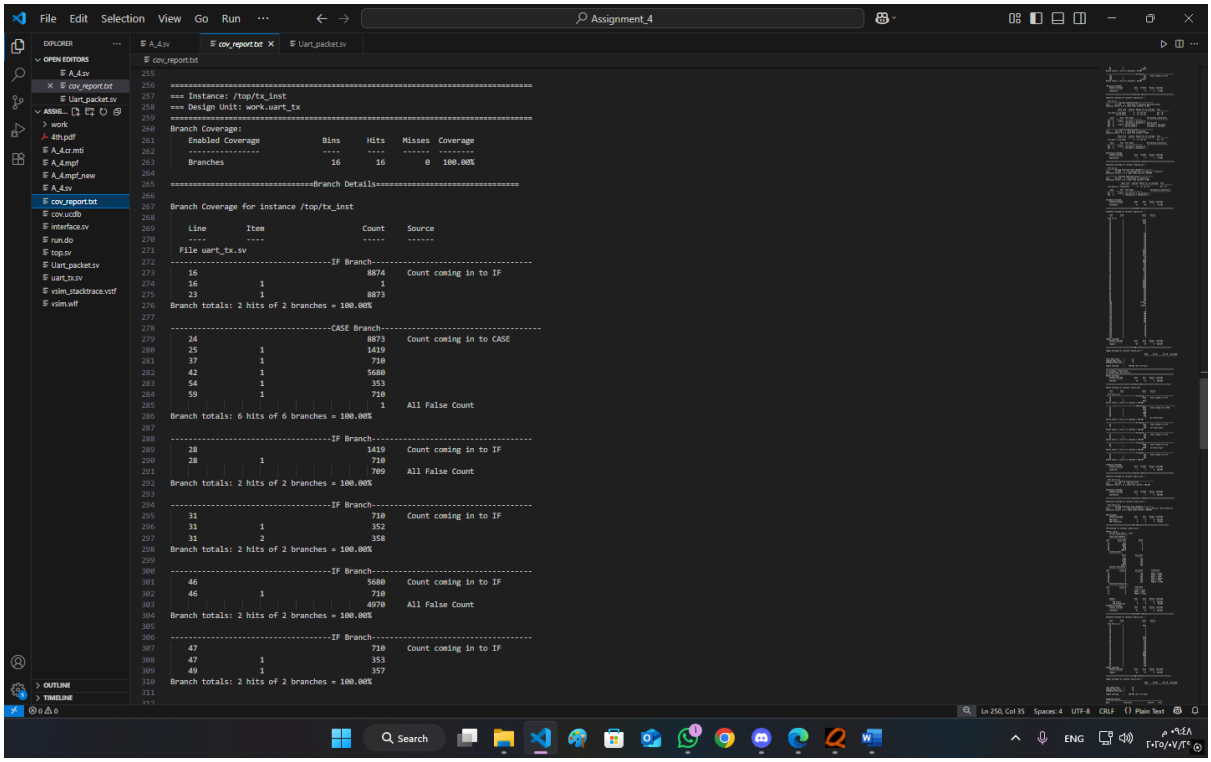




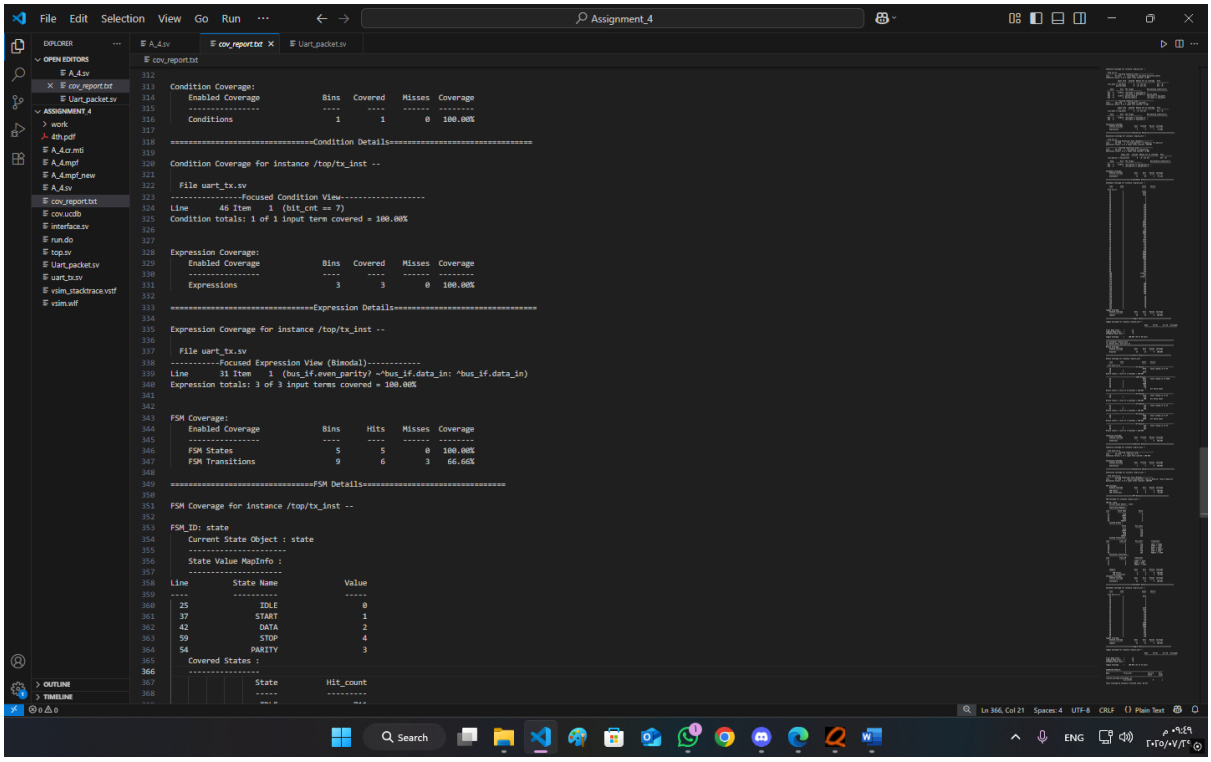


There were a problem with the display statements I couldn't detect

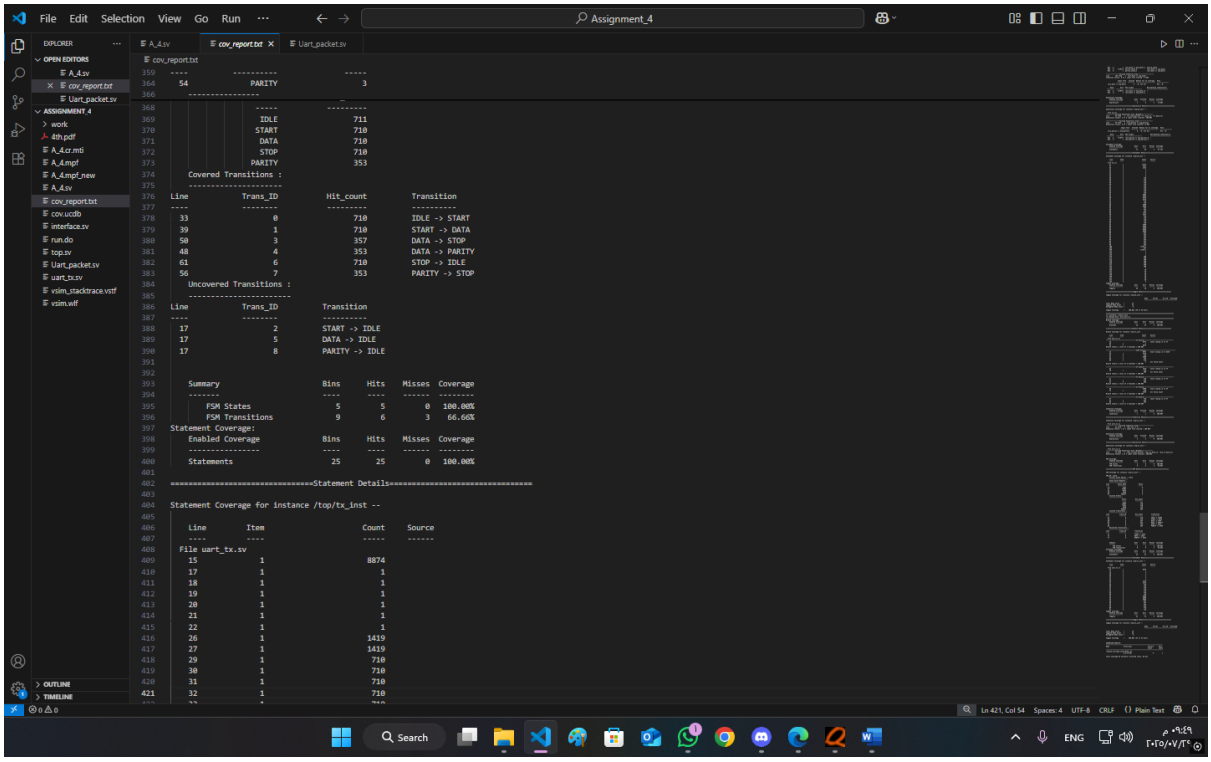
Coverage report snippets:



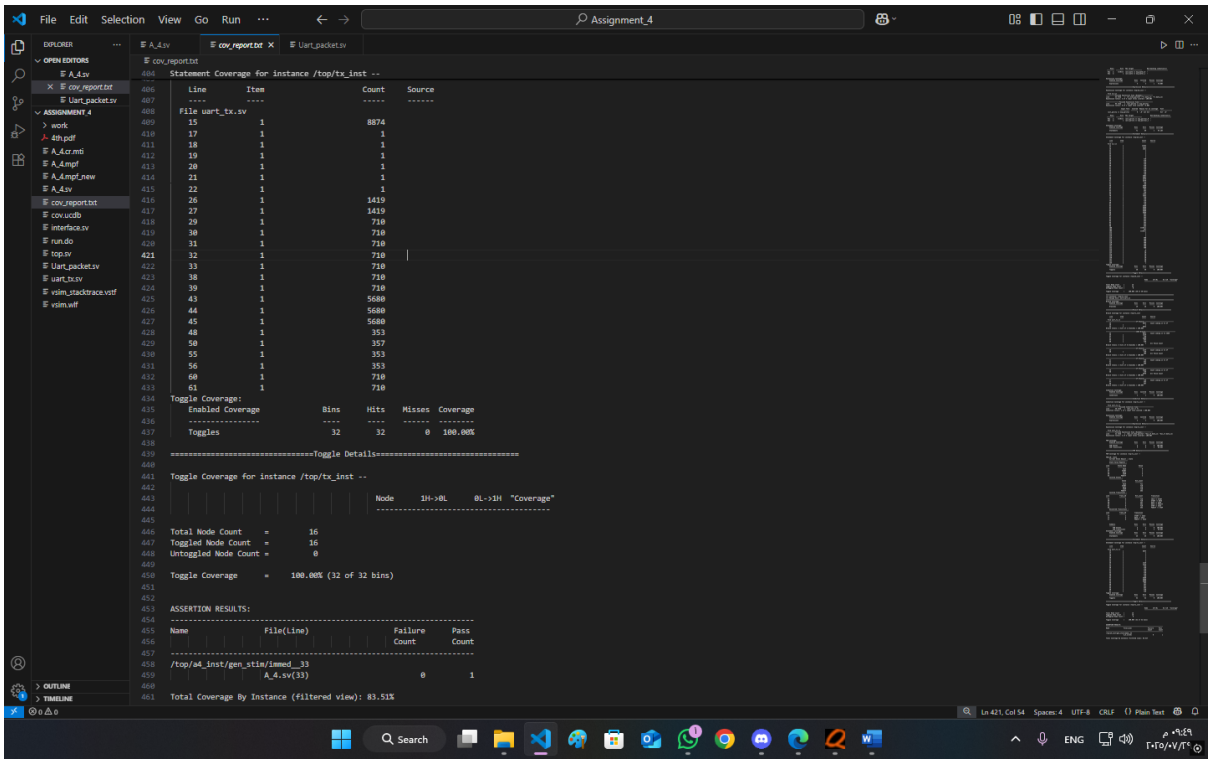
```
==============================================================================
=== Instance: /top/tx_inst
=== Design Unit: work.uart_tx
==============================================================================
Branch Coverage:
    Enabled Coverage              Bins      Hits    Misses  Coverage
    ----------------              ----      ----    ------  --------
    Branches                        16        16         0   100.00%

================================Branch Details================================

Branch Coverage for instance /top/tx_inst

    Line         Item                      Count     Source
    ----         ----                      -----     ------
  File uart_tx.sv
------------------------------------IF Branch------------------------------------
    16                                      8874     Count coming in to IF
    16             1                           1
    23             1                        8873
Branch totals: 2 hits of 2 branches = 100.00%

------------------------------------CASE Branch----------------------------------
    24                                      8873     Count coming in to CASE
    25             1                        1419
    37             1                         710
    42             1                        5680
    54             1                         353
    59             1                         710
                                                1     All False Count
Branch totals: 6 hits of 6 branches = 100.00%

------------------------------------IF Branch------------------------------------
    28                                      1419     Count coming in to IF
    28             1                         710
                                             709     All False Count
Branch totals: 2 hits of 2 branches = 100.00%

------------------------------------IF Branch------------------------------------
    31                                      710     Count coming in to IF
    31             1                         352
    31             2                         358
Branch totals: 2 hits of 2 branches = 100.00%

------------------------------------IF Branch------------------------------------
    46                                      5680     Count coming in to IF
    46             1                         710
                                            4970     All False Count
Branch totals: 2 hits of 2 branches = 100.00%

------------------------------------IF Branch------------------------------------
    47                                      710     Count coming in to IF
    47             1                         353
    49             1                         357
Branch totals: 2 hits of 2 branches = 100.00%
```



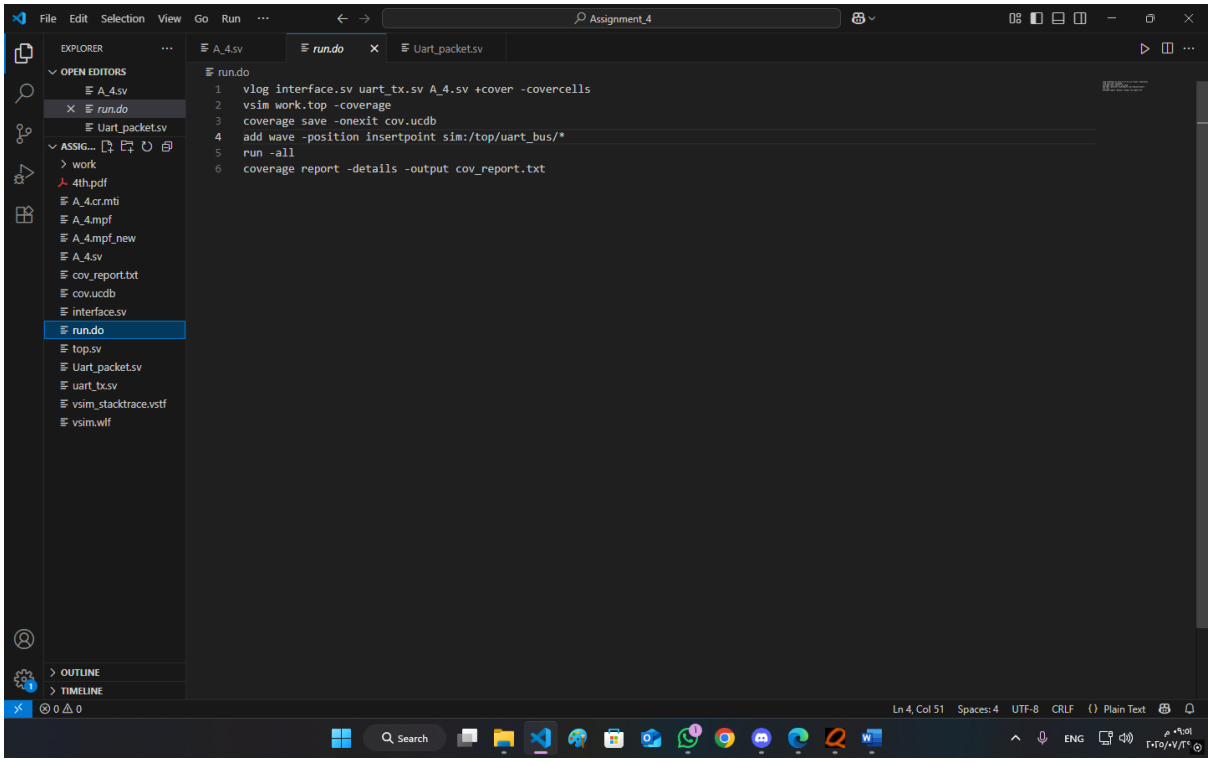
```
Condition Coverage:
    Enabled Coverage              Bins    Covered    Misses  Coverage
    ----------------              ----    ----       ------  --------
    Conditions                       1          1         0   100.00%

================================Condition Details================================

Condition Coverage for instance /top/tx_inst --

  File uart_tx.sv
----------------Focused Condition View-------------------
Line       46 Item    1  (bit_cnt == 7)
Condition totals: 1 of 1 input term covered = 100.00%


Expression Coverage:
    Enabled Coverage              Bins    Covered    Misses  Coverage
    ----------------              ----    ----       ------  --------
    Expressions                       3          3         0   100.00%

================================Expression Details================================

Expression Coverage for instance /top/tx_inst --

  File uart_tx.sv
-----------Focused Expression View (Bimodal)-----------
Line       31 Item    1  (bus_if.even_parity? ~^bus_if.data_in: ^bus_if.data_in)
Expression totals: 3 of 3 input terms covered = 100.00%


FSM Coverage:
    Enabled Coverage              Bins      Hits    Misses  Coverage
    ----------------              ----      ----    ------  --------
    FSM States                       5         5         0   100.00%
    FSM Transitions                  9         6         3    66.66%

================================FSM Details================================

FSM Coverage for instance /top/tx_inst --

FSM_ID: state
    Current State Object : state
    ----------------------------
    State Value MapInfo :
    ---------------------
Line          State Name              Value
----          ----------              -----
  25             IDLE                     0
  37            START                     1
  42             DATA                     2
  59             STOP                     4
  54           PARITY                     3
    Covered States :
    ----------------
                State          Hit_count
                -----          ---------
```



```
 54                PARITY                  3
   ----------------
                   -----            ---------
                    IDLE                711
                   START                710
                    DATA                710
                    STOP                710
                  PARITY                353
   Covered Transitions :
   ---------------------
Line            Trans_ID          Hit_count          Transition
----            --------          ---------          ----------
  33                   0                710          IDLE -> START
  39                   1                710          START -> DATA
  50                   3                357          DATA -> STOP
  48                   4                353          DATA -> PARITY
  61                   6                710          STOP -> IDLE
  56                   7                353          PARITY -> STOP
   Uncovered Transitions :
   -----------------------
Line            Trans_ID          Transition
----            --------          ----------
  17                   2          START -> IDLE
  17                   5          DATA -> IDLE
  17                   8          PARITY -> IDLE


   Summary                       Bins     Hits   Misses  Coverage
   -------                       ----     ----   ------  --------
       FSM States                   5        5        0   100.00%
       FSM Transitions              9        6        3    66.66%
Statement Coverage:
   Enabled Coverage              Bins     Hits   Misses  Coverage
   -----------------             ----     ----   ------  --------
   Statements                      25       25        0   100.00%

=============================Statement Details=============================

Statement Coverage for instance /top/tx_inst --

   Line         Item                      Count     Source
   ----         ----                      -----     ------
 File uart_tx.sv
   15             1                       8874
   17             1                          1
   18             1                          1
   19             1                          1
   20             1                          1
   21             1                          1
   22             1                          1
   26             1                       1419
   27             1                       1419
   29             1                        710
   30             1                        710
   31             1                        710
   32             1                        710
```



```
Statement Coverage for instance /top/tx_inst --

   Line         Item                      Count     Source
   ----         ----                      -----     ------
 File uart_tx.sv
   15             1                       8874
   17             1                          1
   18             1                          1
   19             1                          1
   20             1                          1
   21             1                          1
   22             1                          1
   26             1                       1419
   27             1                       1419
   29             1                        710
   30             1                        710
   31             1                        710
   32             1                        710
   33             1                        710
   38             1                        710
   39             1                        710
   43             1                       5680
   44             1                       5680
   45             1                       5680
   48             1                        353
   50             1                        357
   55             1                        353
   56             1                        353
   60             1                        710
   61             1                        710
Toggle Coverage:
   Enabled Coverage              Bins     Hits   Misses  Coverage
   -----------------             ----     ----   ------  --------
   Toggles                         32       32        0   100.00%

=============================Toggle Details=============================

Toggle Coverage for instance /top/tx_inst --

                                          Node      1H->0L      0L->1H  "Coverage"
                                          ----------------------------------------


Total Node Count     =        16
Toggled Node Count   =        16
Untoggled Node Count =         0

Toggle Coverage      =    100.00% (32 of 32 bins)


ASSERTION RESULTS:
--------------------------------------------------------------------
Name                File(Line)                    Failure      Pass
                                                  Count        Count
--------------------------------------------------------------------
/top/a4_inst/gen_stim/immed__33
                    A_4.sv(33)                         0          1

Total Coverage By Instance (filtered view): 83.51%
```

The FSM coverage problem can’t be solved for the FSM have only one sequence of operation thus it can’t make all transitions.

DO file snippets: